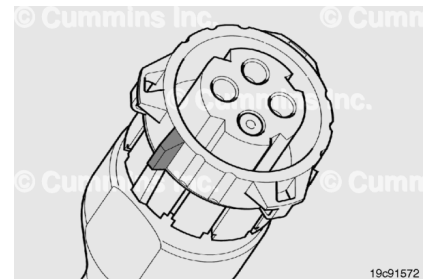


Trainings ▾

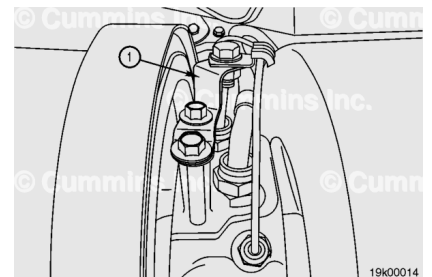
Remove

Disconnect the sensor at its electrical connector.

Inspect the electrical connector and pins for signs of damage.



Remove and discard the wiring retainer bracket (1) and hardware, if equipped. This bracket is no longer used to route the turbocharger speed sensor wire.

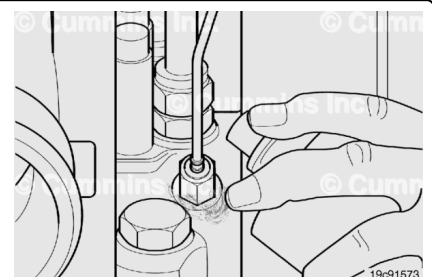


⚠ WARNING ⚠

To reduce the possibility of personal injury, wear goggles and protective clothing.

Note : Make sure the speed sensor and area directly around it is free of debris before removal to make sure no foreign material enters the turbocharger bearing housing as the sensor is removed.

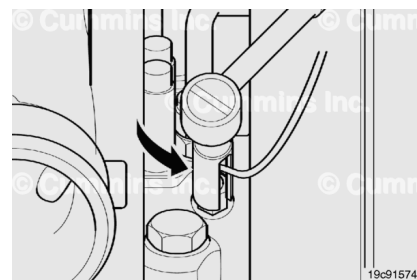
Where road spray, dirt, and fluid ingress has caused fouling or corrosion of the speed sensor, apply a penetrating oil in accordance with the oil manufacturers instructions.



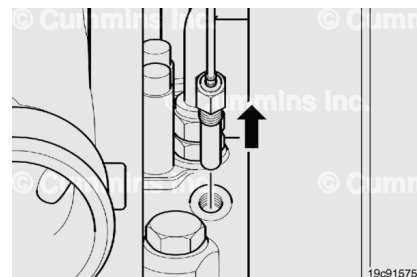
Use a 13 mm, 6 point, flare nut sensor socket, or equivalent, to loosen the speed sensor.

Note : Breaking the sealed joint may require significant torque. It is essential that the sensor socket is properly

engaged on the sensor to avoid inadvertent damage to the cable and connector.

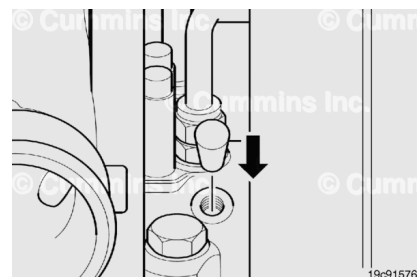


Remove the speed sensor and place on a clean surface to avoid collection of unwanted ferrous particles.



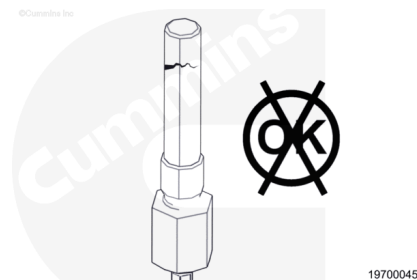
Note : Removal of speed sensor leaves an open access to the bearing housing. It is essential that no dirt or fluids enter the bearing housing cavity during speed sensor replacement.

Insert clean rubber plug.



Inspect for Reuse

Inspect the turbocharger speed sensor body for signs of cracking, contact marks, or other damage.

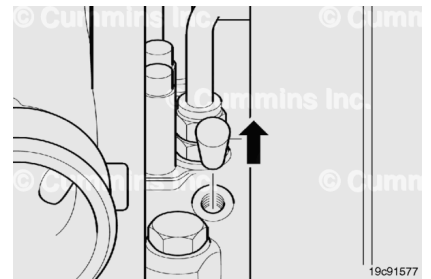


Install

Note : Always wear safety glasses.

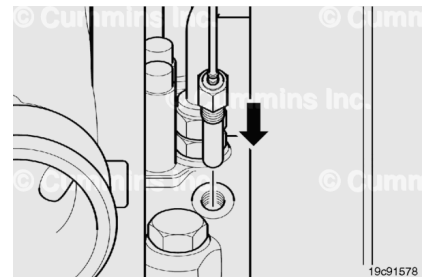
Remove the rubber plug.

Inspect speed sensor seat area of the turbocharger bearing housing. Make sure it is clean and free of corrosion or deformities. This will make sure of a good seal.



Prior to inserting speed sensor, make sure it is clean and unable to collect magnetic particles as it is installed.

The sensor is designed to seal directly in the bearing housing without any supplementary sealing. Do **not** install an o-ring or sealing washer.

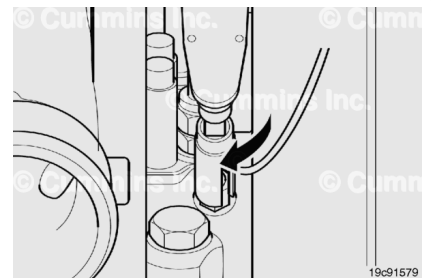


Note : Always use a special socket wrench to prevent damage to sensor cable. Use a 13 mm, 6 point, flare nut socket, or equivalent, to install and torque the sensor.

Note : Accurate torque wrench selection and calibration is critical to make sure of a leak-proof compression joint between sensor and bearing housing.

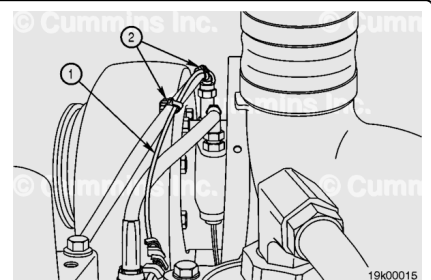
Tighten the speed sensor.

Torque Value: 12 n•m [106 in-lb]



⚠ CAUTION ⚠

The turbocharger speed sensor harness must be tied up and away from any heat sources, and hard or sharp surfaces. Failure to do so will cause damage to the harness.



⚠ CAUTION ⚠

Do not install cable ties directly over the sensor connector. Doing so can damage the connector. Tie the speed sensor wiring harness on either side of the connectors.

Use two metal wire tie clips (2) to attach the turbocharger speed sensor wire (1) to the lubricating oil supply line, as illustrated.

Push the sensor electrical connectors together until they lock.

Note : The connector must be fully mated.

Start the engine and verify with INSITE™ electronic service tool that the turbocharger 1 speed sensor is working properly.

Last Modified: 20-Jul-2015
